6.4.6 One parameter families

(i) **when m is arbitrary and b is a fixed constant**

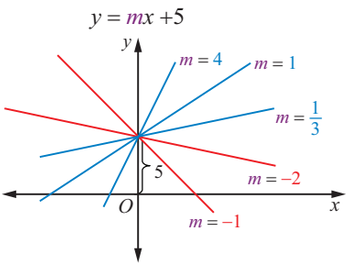


Figure 6.40

Let us find the family of equations of straight lines for the line $y = m\,x+b$ by considering m is arbitrary constant and b is a fixed constant say $b = 5$. Therefore the equation, for different real values of m, represents a family of lines with y-intercept 5 units. A few members of this family are shown in figure. For example, in this diagram the slope m takes $-1,\ -2,\ \frac{1}{3},\ 1$ and 4

Example 6.28 Find the equations of the straight lines in the family of the lines $y = mx + 2$, for which m and the x-coordinate of the point of intersection of the lines with $2x+3y = 10$ are integers.

Solution:

To find the equations of straight lines for the family of line $y = mx + 2$, we have to determine the values of the parameter m.

The point of intersection of the lines $y = mx + 2$ and $2x + 3y = 10$ is

$$\left(\frac{4}{3m+2}, \frac{10m+4}{3m+2}\right)$$

It is given that the slope m and the x-coordinate are integers.

$\therefore\ \dfrac{4}{3m+2}$ is an integer $\Rightarrow 3m + 2$ is a divisor of 4 $(\pm 1,\ \pm 2$ and $\pm\, 4)$

$\therefore\ 3m + 2 = \pm\,1,\quad 3m + 2 = \pm\,2,\quad 3m + 2 = \pm\,4$, where m is an integer

Solving we get, $m = \{-2, -1, 0\}$

The equations are, $y = -2x + 2$, $y = -x + 2$ and $y = 2$

(ii) **when b is arbitrary and m is a fixed constant**.

As discussed above, suppose b is arbitrary constant and m is a fixed constant say $m = -2$,the equation $y = m\,x + b$ becomes $y = -2x + b$. For different real values of b, a family of lines can be obtained with slope -2. A few members of this family are shown in the figure. For example, in this diagram b can take values $-3,\ -1,\ 0,\ 1,\ 2,\ 3$ and 4.

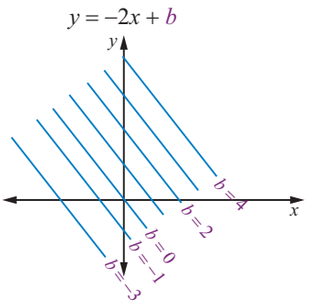


Figure 6.41

Two special cases family of parallel lines and family of perpendicular lines are given below